

Zeyu Chen (Jeffrey)

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EDUCATION

University of Michigan, Ann Arbor

May 2026 (Expected)

Master of Science in Robotics Engineering; GPA: 3.93/4.00

Ann Arbor, MI

Coursework: Self Driving Cars, Mobile Robotics, Deep Learning for Robot Perception, Action and Perception.

University of California, San Diego

Mar 2023

Bachelor of Science in Mechanical Engineering (Control and Robotics); GPA: 3.74/4.00

La Jolla, CA

Coursework: Motion Planning & Estimation, Data Structures, Control Systems, Machine Learning, Mechanical Design.

SKILLS

Programming Languages & Software: Python, C/C++, Java, Unix Scripting, MATLAB, SolidWorks, Microsoft Office

Robotics Tools: ROS/ROS 2, Linux, Docker, Git, GitHub, GitLab, Jira, PyTorch, OpenCV, State Machines, PyBullet

Certifications: NVIDIA Jetson AI Specialist Certificate, Certified SolidWorks Associate (CSWA) – Mechanical Design

EXPERIENCE

Field Robotics Group (FroG)

Feb 2026 – Present

Graduate Researcher — Robotics Software

Ann Arbor, MI

- Developing a **ROS 2** software stack for an autonomous underwater vehicle (AUV) in **Python** and **C++**, from high-level wrench command generation through actuator control allocation.
- Integrated the stack with Pixhawk hardware running ArduSub via a MAVROS2-based **MAVLink** interface.
- Dockerized** the workspace and automated bring-up with shell scripts for repeatable development and deployment.

Fluent Robotics Lab

Jan 2025 – Present

Graduate Researcher

Ann Arbor, MI

- Building an end-to-end human-in-the-loop system in **ROS Noetic**, enabling interactive task execution via **LLM-based** planning with natural-language commands, object perception, and a motion-planning backend for object arrangement.
- Deployed an **RGB-D** perception pipeline combining **visual language models**, **YOLO-based** segmentation, **OpenCV** processing, and **point cloud** alignment for real-time object pose estimation.

Symbiotic Inc.

May 2025 – Aug 2025

Robotics Software Intern

Wilmington, MA

- Engineered autonomous “Smart Move” routines in **C++** and **Python**, integrated into existing robot recovery systems to enhance reliability and efficiency across large-scale teleoperation and case-handling workflows in warehouse automation.
- Collaborated cross-functionally to collect downstream requirements, design and test Smart-Move features, and deploy final software updates, reducing operational time by **42%–80%** with **98%–100%** task success rates.

Multi-Agent Robotics (MURO) Lab

Apr 2023 – Jul 2024

Undergraduate Researcher

La Jolla, CA

- Implemented a multi-robot localization system using **ROS 2**, Python, and sensor fusion techniques (including LiDAR, IMU, camera, etc.) for real-time global pose estimation.
- Customized a **Multiplicative Extended Kalman Filter (MEKF)** to mitigate the position and orientation errors of moving robots, improving real-time positioning accuracy by **>70%**.
- Co-developed a distributed localization algorithm with uncertainty quantification for robust object position estimation under sensor faults.

Existential Robotics Laboratory

Jan 2023 – Jun 2024

Undergraduate Researcher

La Jolla, CA

- Engineered an interactive robotics learning platform integrating robot navigation algorithms and visualizations with **Python**, **Google Colab**, and the **PyBullet** simulation tool, enabling learning and rapid prototyping of robot localization, mapping, path planning, and motion control.
- Contributed to a centralized [web-based simulation interface](#) with sample implementations and interactive visualizations, reducing access time to testing tools by **>90%** and streamlining online algorithm evaluation.

PROJECTS

MBot Navigation Autonomy

Oct 2024 – Dec 2024

Robotics Software Engineer | Technologies: C/C++, Python, SLAM, LCM Message-Passing Framework, 3D Printing

- Built an autonomous mobile robot with advanced navigation capabilities leveraging simultaneous localization and mapping (SLAM), path planning, motion control, and sensor fusion.
- Implemented core algorithms, including PID controllers, Particle Filter, and Occupancy Grid Mapping algorithms in C/C++, achieving reliable indoor autonomy.